

Nikhil VS

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[LinkedIn](#) | [Leetcode](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Final-year Information Science undergraduate (East West Institute of Technology, 2027) with 2 deployed AI/ML products and Amazon HackOn With Amazon 6.0 Round 1 clearance. Developed a Fraud Detection System achieving 97.92% ROC-AUC using XGBoost with SHAP explainability, deployed on cloud. Built an AI Code Review platform integrating local LLMs with FastAPI and Docker. Completed Andrew Ng's Machine Learning Specialization. Strong foundation in probability, statistics, and linear algebra applied to ML model development. Seeking ML internship to apply machine learning, model development, and software engineering skills at scale.

SKILLS

- **ML/AI:** XGBoost, Scikit-learn, SHAP Explainability, NumPy, Pandas, Matplotlib, Seaborn, LLMs, Ollama, Prompt Engineering, Supervised Learning, Classification, SMOTE, Probability & Statistics, Linear Algebra, Feature Engineering, Cross-Validation, Ensemble Methods, Neural Networks, Deep Learning (basics)
- **Backend:** FastAPI, Flask, SQLAlchemy, Alembic, JWT, OAuth2, Pydantic, REST APIs
- **Frontend:** React, Vite, TypeScript, Axios, Chart.js, Tailwind CSS
- **Databases:** PostgreSQL, Neon, SQLite
- **DevOps:** Docker, Docker Compose, Git, GitHub, Render, Vercel
- **Languages:** Python, JavaScript, TypeScript, SQL, C++
- **Core CS:** DSA, OOP, DBMS, Operating Systems, Algorithm Analysis, Data Structures

PROJECTS

Fraud Detection System

[GitHub: <https://github.com/Nikhil-VS1811/fraud-detection-system>] [Live Demo: <https://fraud-frontend-vxzh.onrender.com>]

Tech: XGBoost, FastAPI, React, PostgreSQL, SHAP, Docker, Neon, Render

- Built end-to-end ML pipeline on 284,807 real transactions, achieving 97.92% ROC-AUC and 80% F1-Score using XGBoost classifier
- Addressed 1:579 class imbalance using SMOTE oversampling, improving fraud recall to 89%
- Implemented SHAP explainability layer exposing top feature contributors per prediction rendered in React dashboard
- Designed RESTful FastAPI backend with 6 endpoints, SQLAlchemy ORM, and PostgreSQL prediction logging
- Deployed full-stack: React frontend on Render, FastAPI backend on Render, PostgreSQL on Neon cloud

AI Code Review System

[GitHub: <https://github.com/Nikhil-VS1811/AI-Code-Reviewer>]

Tech: FastAPI, React, TypeScript, PostgreSQL, LLM, Ollama, Docker, JWT

- Built AI-powered code review platform integrating local LLMs (Ollama/qwen2.5-coder:7b) via clean provider abstraction layer
- Implemented complete auth system: JWT + OAuth2, protected routes, SQLAlchemy ORM with Alembic migrations
- Designed submission and review pipeline with duplicate prevention, AI response validation, and score normalization
- Containerized full stack using Docker Compose (React + FastAPI + PostgreSQL)

CERTIFICATIONS

- Andrew Ng Machine Learning Specialization — Coursera
- Oracle Cloud Infrastructure 2025 — Generative AI Professional
- Oracle Cloud Infrastructure 2025 — Data Science Professional
- Python Programming — Google

EDUCATION

East West Institute of Technology, Bengaluru

B.E. Information Science | 2023 – 2027 | CGPA: 8.88 (up to 5th Semester)

Carmel PU College, Bengaluru

Pre-University | 2021 – 2023 | 94.6%

ACHIEVEMENTS

- Selected for HackOn With Amazon 6.0 — Cleared Round 1
- Selected contributor for GirlScript Summer of Code (GSSoC) 2026
- Solved 250+ DSA problems on LeetCode and Geeks for Geeks
- Studying probability distributions, Bayesian inference, and linear algebra for ML applications